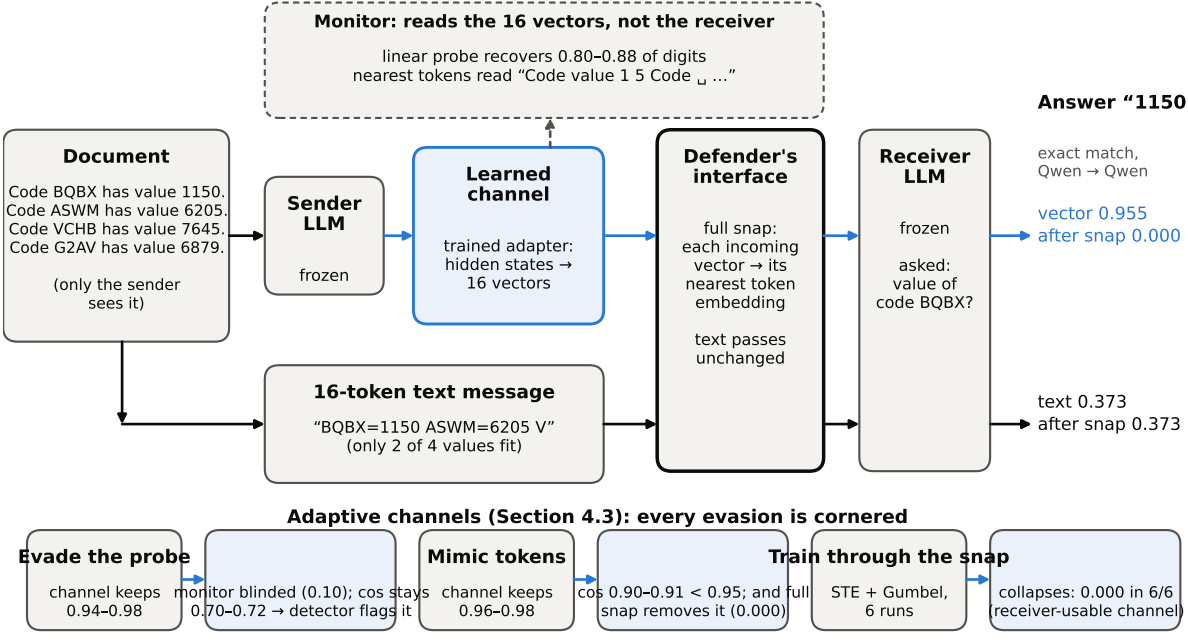
Monitor: reads the 16 vectors, not the receiver
linear probe recovers 0.80–0.88 of digits
nearest tokens read “Code value 1 5 Code ␣ ...”
Document
Code BQBX has value 1150.
Code ASWM has value 6205.
Code VCHB has value 7645.
Code G2AV has value 6879.
(only the sender sees it)
Sender LLM
frozen
Learned channel
trained adapter: hidden states → 16 vectors
Defender's interface
full snap: each incoming vector → its nearest token embedding
text passes unchanged
Receiver LLM
frozen
asked: value of code BQBX?
Answer “1150”
exact match, Qwen → Qwen
vector 0.955
after snap 0.000
16-token text message
“BQBX=1150 ASWM=6205 V”
(only 2 of 4 values fit)
text 0.373
after snap 0.373
Adaptive channels (Section 4.3): every evasion is cornered
Evade the probe
channel keeps 0.94–0.98
monitor blinded (0.10); cos stays 0.70–0.72 → detector flags it
Mimic tokens
channel keeps 0.96–0.98
cos 0.90–0.91 < 0.95; and full snap removes it (0.000)
Train through the snap
STE + Gumbel, 6 runs
collapses: 0.000 in 6/6 (receiver-usable channel)